Structural Balance Dynamics at Nonzero Temperature

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Introduction

This project examines the mean-field treatment of structural-balance dynamics at nonzero temperature developed by Rabbani, Shirazi, and Jafari [1]. It reviews the theoretical background, explains the Hamiltonian formulation and its self-consistency equations, and compares selected published results with simulations carried out for this course project. A related study of quartic balance [2] is considered to discuss the stability of solutions when higher-order interactions are included.

Structural-balance theory describes signed networks containing both positive and negative relationships. Although a balanced configuration contains no frustrated triads, real social networks continue to exhibit tension. Rabbani et al. introduce a finite-temperature extension of balance dynamics in which Boltzmann–Gibbs statistical mechanics assigns an energy to each triad, while temperature represents stochasticity and tolerance of tension. Their mean-field analysis predicts a discontinuous transition between a disordered state and structural balance. Above the model's critical temperature, the ordered balanced branch is absent. Below it, the existence of competing solutions makes the outcome sensitive to initial conditions and the system's evolution [1].

Background on structural balance

Balance theory in signed networks

Balance theory originated in social psychology and provides a framework for studying networks whose relationships may be friendly or hostile. Nodes can represent individuals, groups, or other interacting entities; positive edges represent friendly relationships and negative edges represent hostile ones. The central idea is that people tend to adjust their relationships to reduce cognitive inconsistency and maintain coherent patterns of association.

A triad consists of three nodes and the three edges connecting them. It is balanced when the product of its edge signs is positive. This occurs in two configurations: all three edges are positive, or one edge is positive, and the other two are negative. Conversely, a triad is imbalanced when this product is negative, as occurs with one negative edge and two positive edges, or with three negative edges.

For example, consider three individuals, A, B, and C. A triad in which all three pairs are friends is balanced. A configuration in which A and B are friends but both are hostile toward C is also balanced. If A is friends with B, B is friends with C, and A is hostile toward C, the triad is imbalanced. Balance dynamics describes possible changes in edge signs that reduce such inconsistency.

In a complete signed network, structural balance requires every triad to be balanced. A network containing imbalanced triads may evolve toward balance as its relationships change, although

convergence depends on the updating rule and the noise level. Mathematical models and numerical algorithms can be used to investigate these dynamics.

Applications considered in the original project include the formation of social groups and alliances, international cooperation and conflict, organizational relationships and conflict resolution, and biological or ecological interactions involving cooperation and competition. In each setting, a signed-network representation offers a way to examine how patterns of positive and negative relationships influence collective organization.

Historical development

Fritz Heider formulated balance theory in the 1940s to explain how people perceive interpersonal relationships and seek consistency among their attitudes. The theory concerns configurations of three entities and the positive or negative relations among them. Inconsistent configurations can create psychological discomfort and motivate a change in attitudes or relationships.

Heider's work provided a foundation for subsequent studies of social networks and structural balance. Later developments extended the approach to questions about group formation, conflict, alliances, and the evolution of relational structures. These developments connect social-psychological ideas with network science, sociology, and organizational analysis.

Structural balance and collective organization

Structural balance concerns the organization of relationships across a network, rather than the consistency of a single triad. Balanced triads, coherent sign patterns, and reduced frustration are closely related aspects of the same network-level condition. A triad that could become balanced after an edge change is not yet balanced; structural balance refers to the configuration that is actually present.

For complete networks, balanced configurations can consist of a single friendly group or two groups with positive relationships within each group and negative relationships between them. Thus, structural balance does not imply that every relationship is friendly. An antagonistic division into two internally cohesive groups can satisfy the balance condition.

Finite temperature model

The model studied in [1] examines how stochasticity affects signed social networks through interactions among triads. In earlier balance dynamics, an edge may be selected at random and its sign changed when doing so reduces imbalance. Such a rule tends to favor balanced configurations but excludes energy-increasing moves.

The finite-temperature extension permits energy-increasing changes with a nonzero probability. This represents variation in the tolerance of inconsistent or tense relationships. A Boltzmann factor determines the likelihood of accepting an unfavorable change at a given model temperature.

Temperature T is a control parameter for stochasticity, not a physical temperature measured in degrees Celsius. At high temperature, the statistical preference for lower-energy configurations weakens, and triad configurations approach their random-network frequencies. At low temperature, energetically favorable balanced configurations are strongly preferred, although the trajectory may remain sensitive to

metastability and initial conditions. The model's Hamiltonian and mean-field approximation make it possible to investigate the transition between these regimes.

Hamiltonian and mean field formulation

Triadic Hamiltonian

The analysis considers a fully connected network with N nodes. The sign of an undirected edge is represented by $S_{ij} = S_{ji} \in \{-1, +1\}$. The Hamiltonian is

$$H = -\sum_{i>j>k} S_{ij}\, S_{jk} S_{ki}. \qquad (1)$$

A balanced triad has sign product $+1$ and contributes -1 to the energy; an imbalanced triad contributes $+1$. Consequently, minimizing the Hamiltonian favors structural balance.

To calculate the mean value of an edge, the Hamiltonian is separated into terms involving that edge and a remainder independent of it:

$$H = H_{ij} + H', \qquad H_{ij} = -S_{ij} \sum_{k\neq i,j} S_{jk}\, S_{ki}. \qquad (2)$$

Using the Boltzmann distribution gives the conditional average

$$\langle S_{ij} \rangle_{\text{cond}} = \tanh\left(\beta \sum_{k\neq i,j} S_{jk}\, S_{ki} \right), \qquad \beta = \frac{1}{T}. \qquad (3)$$

The full ensemble average additionally averages over the remaining edge configurations. Under the mean-field closure, the neighboring edge products are replaced by their mean value. Defining $p = \langle S_{ij} \rangle$ and $q = \langle S_{ik} S_{kj} \rangle$ yields

$$p = \tanh[\beta(N-2)q]. \qquad (4)$$

Two edge correlation

To obtain a second self-consistency relation, consider two edges sharing a node. The part of the Hamiltonian involving these edges is

$$H_{ik,kj} = -S_{ik} \sum_{l\neq i,j,k} S_{il}\, S_{lk} - S_{kj} \sum_{l\neq i,j,k} S_{kl}\, S_{lj} - S_{ik} S_{kj} S_{ji}. \qquad (5)$$

Evaluating the four possible sign combinations and applying the mean-field approximation leads to

$$q = \frac{e^{2\beta(N-3)q} - 2e^{-2\beta p} + e^{-2\beta(N-3)q}}{e^{2\beta(N-3)q} + 2e^{-2\beta p} + e^{-2\beta(N-3)q}}. \qquad (6)$$

Equations (4) and (6) form a coupled system for p and q. Substituting the expression for p into the right-hand side of Eq. (6) defines a scalar self-consistency function $f(q; T, N)$. The solutions are the intersections of this function with the identity line $f(q) = q$.

The graphical analysis in [1] shows a single disordered solution at high temperature and, within the coexistence regime below the critical temperature, three solutions: two stable branches separated by an unstable one. The ordered branch approaches $q = 1$ at low temperature; this limiting value should not be assumed to hold exactly at every finite temperature.

Mean triad energy

The mean triad product is $r = \langle S_{ik} S_{kj} S_{ji} \rangle$, and the mean energy per triad is $\bar{E} = -r$. Introducing $a = \beta(N-3)q$ as a shorthand, the expression reproduced in the original report can be written as

$$r = \frac{e^{3a+\beta} - 3e^{a-\beta} + 3e^{-a+\beta} - e^{-3a-\beta}}{e^{3a+\beta} + 3e^{a-\beta} + 3e^{-a+\beta} + e^{-3a-\beta}}. \qquad (7)$$

This quantity distinguishes a balanced configuration, for which $\bar{E} = -1$, from a disordered configuration with approximately equal fractions of balanced and imbalanced triads, for which $\bar{E} \approx 0$.

Bifurcations and stability

A bifurcation diagram describes how the qualitative behavior of a dynamical system changes as a control parameter varies. The control parameter may represent an external influence or a model property, while the state variables describe the system's configuration. Depending on the application, these variables may represent positions, concentrations, population densities, or network observables.

Branches in a bifurcation diagram represent fixed points or other invariant states. A fixed point is a state that remains unchanged under the specified dynamics. Its stability determines whether nearby trajectories approach it or move away from it. Stable and unstable branches are commonly distinguished using solid and dashed lines, filled and open symbols, or different colors.

Examples include pitchfork bifurcations, in which the number and stability of branches change through a symmetry-breaking event; saddle-node bifurcations, in which a pair of fixed points meets and disappears; and transcritical bifurcations, in which two branches exchange stability. More complicated systems may exhibit periodic or chaotic behavior.

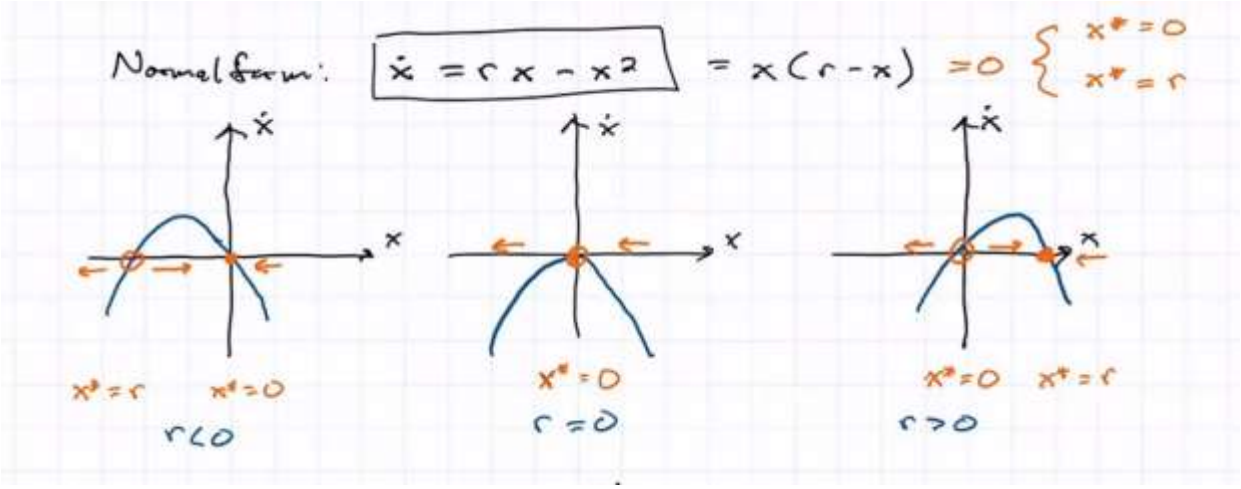


Figure 1. An illustrative transcritical bifurcation diagram retained from the original report. The normal form is $\dot{x} = rx - x^2$, with fixed points $x^* = 0$ and $x^* = r$. The two branches exchange stability as r changes sign. At $r = 0$, the merged fixed point is semistable in the one-dimensional phase-line interpretation. Teaching resources cited in the original report are listed in [3,4].

The finite-temperature balance model is analyzed by locating intersections of $f(q)$ with q. In the original paper, the disappearance of the ordered and unstable branches is described as a "blue-sky" bifurcation [1]. The stable fixed points should be identified through the appropriate dynamics or local stability criterion, rather than by calling them local maxima or minima of the plotted self-consistency curve.

Monte Carlo simulation method

Rabbani et al. test their analytical results using a Metropolis algorithm on a complete network with N nodes [1]. For a fixed temperature, an edge is selected at random and a sign reversal is proposed. The energy difference between the proposed and current configurations is calculated from the Hamiltonian:

$$\Delta E = E_{\text{new}} - E_{\text{old}}. \qquad (8)$$

The proposed change is accepted with probability

$$P_{\text{acc}} = \min\left(1, e^{-\beta \Delta E}\right). \qquad (9)$$

Thus, energy-decreasing moves are accepted, while energy-increasing moves are accepted probabilistically. If the move is rejected, the previous edge sign is retained. Repeating this procedure at different temperatures allows the evolution of mean triad energy and its dependence on temperature to be investigated.

The published analysis compares the analytical branches with Monte Carlo results and identifies a discontinuous change in mean triad energy. Competing stable states and sensitivity to the direction of temperature changes give rise to the hysteresis associated with the first-order transition. A disordered state with mean energy close to zero may persist even where a balanced state is also available.

Analysis of the published results

Rabbani et al. introduce temperature through a Boltzmann–Gibbs description of structural balance and derive a mean-field treatment of the resulting dynamics [1]. The following discussion examines the graphical solutions and the numerical evidence presented in their paper.

Self consistency and the critical temperature

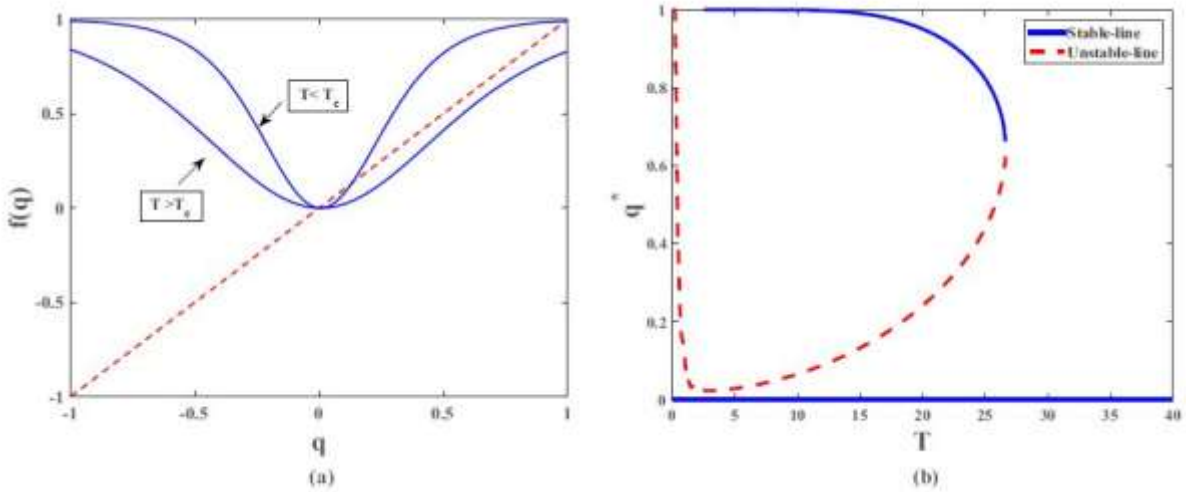


Figure 2. Graphical self-consistency analysis and bifurcation diagram from [1, Fig. 2], retained from the source report. The network has $N = 50$ nodes. The two temperatures shown in the published comparison are $T = 15$ and $T = 28$, below and above the reported critical value $T_c \approx 26.2$.

The intersections in Figure 2(a) show that the number of solutions depends on temperature. Above T_c, a single stable disordered solution remains. In the coexistence regime below T_c, three fixed points are present: two stable solutions and an intermediate unstable solution. Figure 2(b) tracks the corresponding branches as temperature changes and illustrates the abrupt loss of the ordered state at the transition.

Relaxation and limitations of the approximation

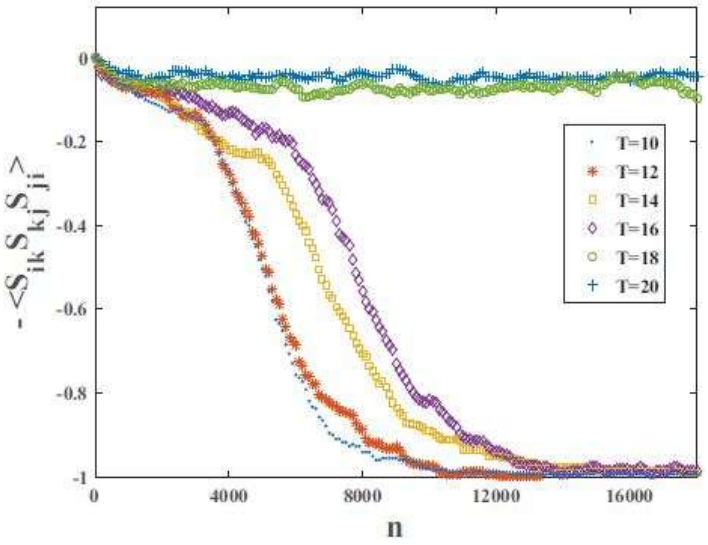


Figure 3. Mean triad energy versus Monte Carlo steps for a complete network with $N = 50$, from [1, Fig. 3]. The temperatures are $T = 10,12,14,16,18$, and 20. The figure is retained from the original report.

At the lower temperatures in Figure 3, the mean energy decreases toward the balanced value. At $T = 18$ and $T = 20$, the trajectories remain close to the disordered branch over the displayed simulation interval. These observations indicate temperature-dependent relaxation and the persistence of competing states. They do not, by themselves, demonstrate that the numerical method or the mean-field approximation has failed.

The original project discussion considered the limitations of mean-field theory near phase transitions. Such approximations replace fluctuating local environments with averaged quantities and can neglect correlations that are important for collective behavior. Four issues are relevant when evaluating their range of validity: critical exponents, the behavior of correlations, scaling relations, and fluctuation effects. Their importance depends on the interaction structure, effective dimensionality, system size, and the nature of the transition.

In continuous transitions, fluctuations may alter mean-field critical exponents, and a correlation length can become large or diverge. Mean-field theory can itself predict a divergent correlation length and can satisfy some scaling relations, so these properties should not be described as universally absent from mean-field treatments. More detailed methods, including renormalization-group approaches where applicable, account for fluctuation effects beyond a simple mean-field closure.

The present model concerns a finite complete network and a first-order transition. General arguments about spatial correlation lengths and continuous-transition exponents cannot establish the cause of the behavior in Figure 3. For this problem, metastability, the observation time, finite-size effects, and the details of the approximation must be examined before attributing discrepancies to neglected fluctuations. The paper also discusses incomplete sampling of metastable states as a possible source of disagreement between analytical and simulation transition points [1].

Analytical and numerical energy branches

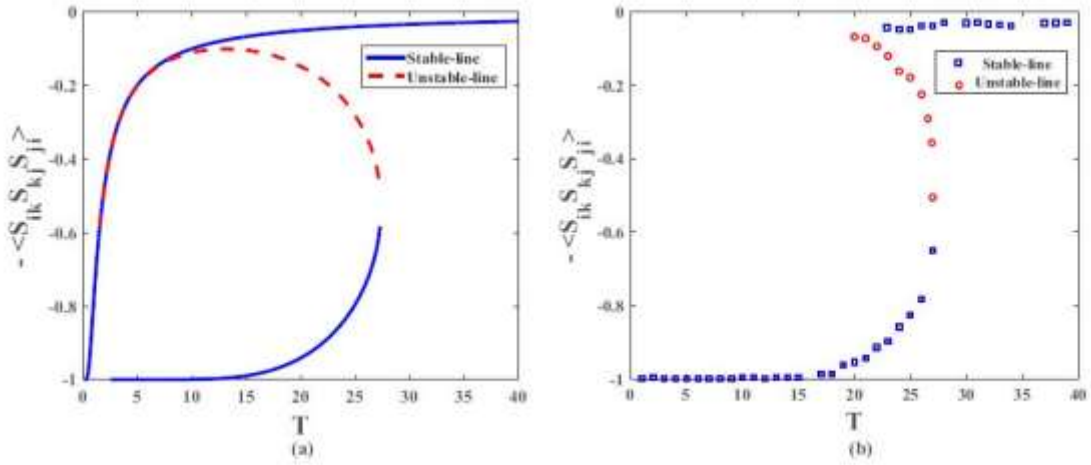


Figure 4. Analytical mean triad-energy branches and Monte Carlo results from [1, Fig. 4]. The solid blue and dashed red curves distinguish stable and unstable analytical branches. The panels are retained from the source report.

The comparison in Figure 4 shows that the simulations recover the broad structure of the theoretical energy branches and the discontinuous transition. The original report identifies $T_c \approx 26.2$ as the reference mean-field value. Agreement in overall shape should be distinguished from an exact numerical match of the transition temperature, particularly in a region affected by metastability and finite observation times.

Dependence on initial configuration

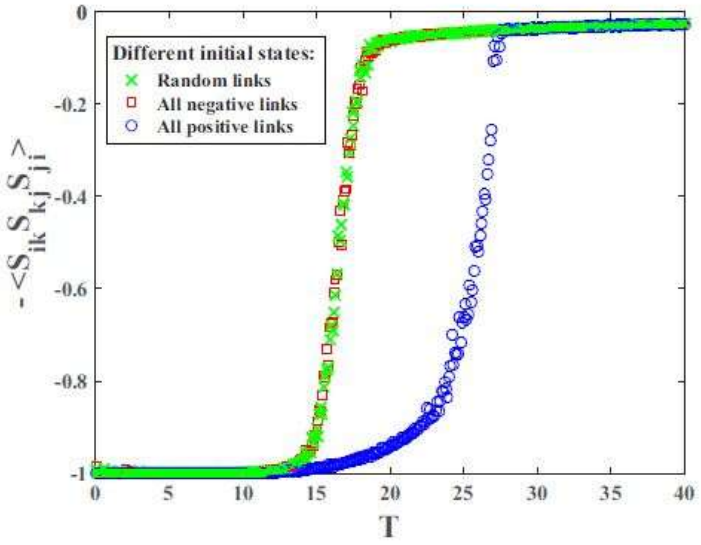


Figure 5. Mean triad energy as a function of temperature for all-positive, all-negative, and random initial edge configurations, from [1, Fig. 5]. The figure is retained from the source report.

The all-positive initial configuration remains on the low-energy balanced branch up to a temperature near the reported critical value. The all-negative and random initial configurations approach the disordered branch at lower temperatures and have similar profiles in the published figure. The comparison

demonstrates that the final observed state depends on the initial configuration and the subsequent trajectory through the coexistence region.

Simulations carried out for this project

Reproducing the graphical self-consistency analysis

I evaluated the self-consistency function on both sides of the reported critical temperature, following the graphical analysis in Figure 2(a) of [1]. The project comparison uses $T = 21.2$ and $T = 31.2$, five model-temperature units below and above $T_c = 26.2$.

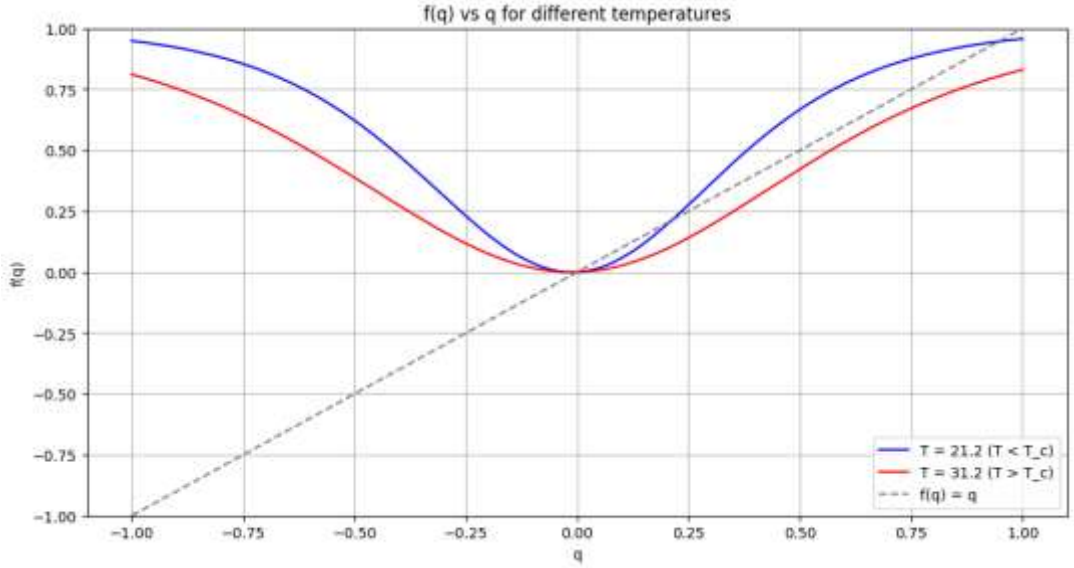


Figure 6. Project calculation of $f(q)$ versus q at $T = 21.2$ and $T = 31.2$. The dashed line is the identity line, and its intersections with each curve identify the self-consistent solutions.

The lower-temperature curve has three intersections, whereas the higher-temperature curve has one. This reproduces the qualitative structure reported in the paper. The two outer solutions in the coexistence region correspond to the stable branches, while the intervening solution is unstable under the stability interpretation used in the published analysis.

Bifurcation diagram

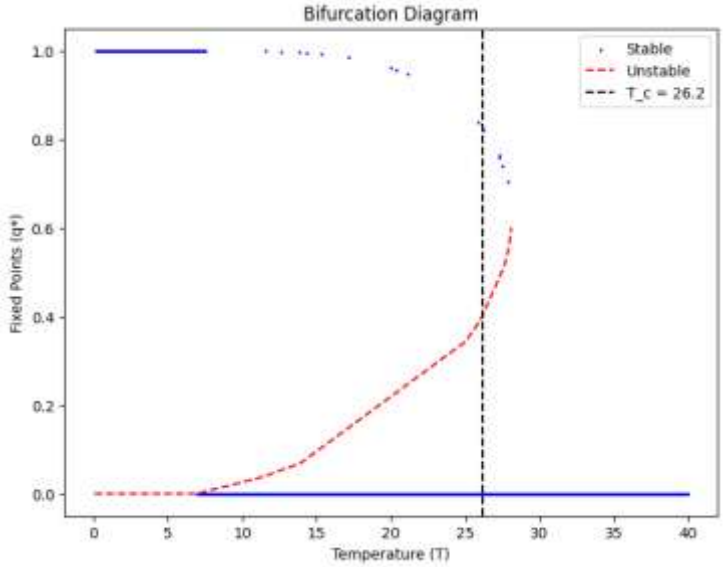


Figure 7. Bifurcation diagram obtained for this project. Stable and unstable branches are identified by blue and red markers. The dashed vertical line marks the published reference value $T_c = 26.2$; it should not be read as an independently extracted transition temperature.

I also reproduced the branch diagram corresponding to Figure 2(b) of [1]. The resulting diagram captures the coexistence of stable and unstable branches and their disappearance at higher temperature. In the original analysis, the discrepancy in the transition estimate was described as approximately one model-temperature unit. A precise error estimate would require extracting the turning point from the numerical solutions and reporting the temperature grid and solver tolerances.

Below the transition, the shape of the project branch diagram resembles the theoretical curve more closely than the Monte Carlo branch reconstruction shown elsewhere in the paper. This is consistent with the different procedures being compared, but does not establish the quantitative accuracy of the project calculation.

Evolution of mean triad energy

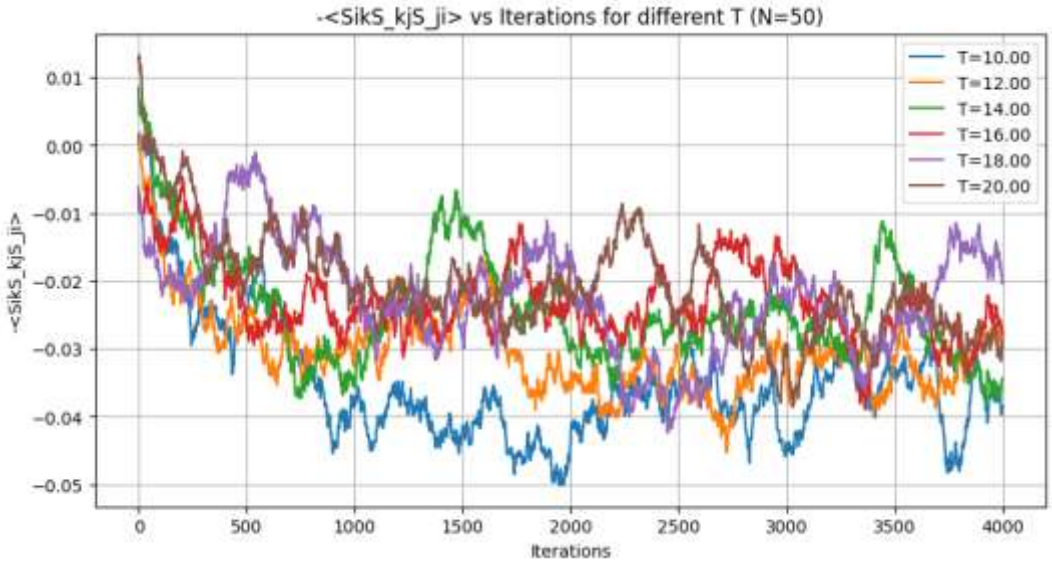


Figure 8. Project traces of mean triad energy for $N = 50$ and $T = 10,12,14,16,18$, and 20, over 4,000 recorded update steps. The time convention should be matched to the published simulation before quantitative comparisons are made.

To compare with Figure 3 of [1], I examined the evolution of mean triad energy at several temperatures. Computational limitations restricted the simulations to 4,000 recorded steps. The traces show an initial tendency toward negative energy but fluctuate close to the disordered value during the available interval; they do not reach the fully balanced value of -1.

The behavior observed in the longer published simulations was not reproduced within this interval. Because the project runs are shorter and the update-time normalization is not specified in the report, this observation cannot establish either agreement or disagreement in the long-time limit. Longer runs and a matched definition of a Monte Carlo step would be required to assess relaxation and metastability reliably.

Initial conditions and temperature dependence

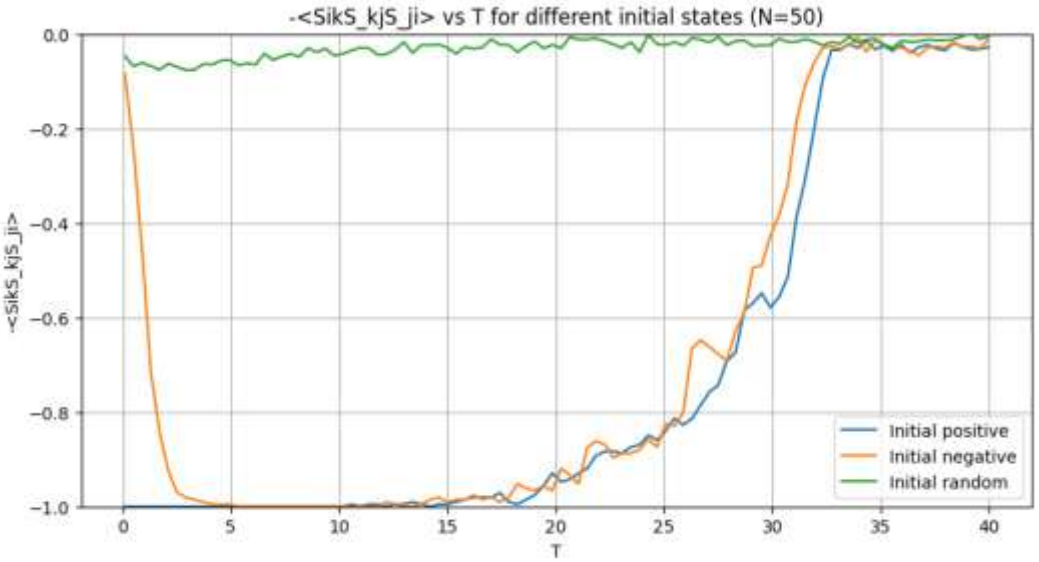


Figure 9. Project mean triad-energy curves for all-positive, all-negative, and random initial configurations. The three curves preserve the output presented in the original report.

Following Figure 5 of [1], I studied the mean triad energy as a function of temperature for three initial configurations. As in the published comparison, the all-positive trajectory generally lies below the all-negative trajectory, and the all-negative trajectory approaches the high-temperature disordered level slightly earlier. However, the random initial configuration in the project calculation remains close to zero with substantial fluctuations, rather than closely following the all-negative curve as it does in the published figure. The separation between the all-positive and all-negative transition regions is also smaller in the project result.

The original analysis identified an apparent displacement of roughly five temperature units toward higher temperature. Shifting the curves to the left would bring their saturation region closer to the reference transition temperature, but this graphical observation does not explain the discrepancy. The temperature convention, energy normalization, equilibration time, and implementation would need to be checked before applying or interpreting such a shift.

Stability in triadic and quartic balance models

Fixed points in the quartic model

The second paper, by Kargaran et al. [2], extends the interaction structure to quartic balance. Its self-consistency analysis uses q, the mean product of two edges sharing a node, and o, the mean product associated with an open square. Intersections of the coupled self-consistency curves determine the fixed points in the (q, o) plane. This section discusses Figures 2 and 3 of that paper; those figures were not embedded in the original project file.

Below the critical temperature, the published analysis shows five fixed points in the illustrated coexistence regime. Three are attractive, shown as blue circles, and two are repulsive, shown as red triangles. A trajectory initialized near an attractive solution approaches it under the specified iteration,

whereas perturbations away from a repulsive solution grow. The stability of the origin also depends on temperature, so the illustrated count of stable solutions should not be generalized to every temperature below the transition.

At the critical temperature, the nontrivial branches meet through tangency. The paper counts three solutions at this threshold, including the origin and the coalescing nontrivial branches. Above the critical temperature, only the stable trivial solution at $q = o = 0$ remains. This solution has vanishing mean two-star and open-square observables and corresponds to the disordered branch in the high-temperature regime.

The coexistence of three stable solutions is more accurately described as multistability or tristability than bistability. The ordered branches are associated with the all-positive and all-negative configurations, termed “heaven” and “hell” in [2], while the origin can also represent a random or bipolar configuration at the level of these observables. Quartic balance and triadic structural balance impose different conditions: an all-negative network can be favorable under a quartic Hamiltonian while all of its triangles remain imbalanced under the triadic sign-product criterion.

Illustrative vector fields for the triadic model

The original report also includes vector-field plots intended to illustrate the stability of the first paper’s mean-field solutions. The plots are retained below in their original form.

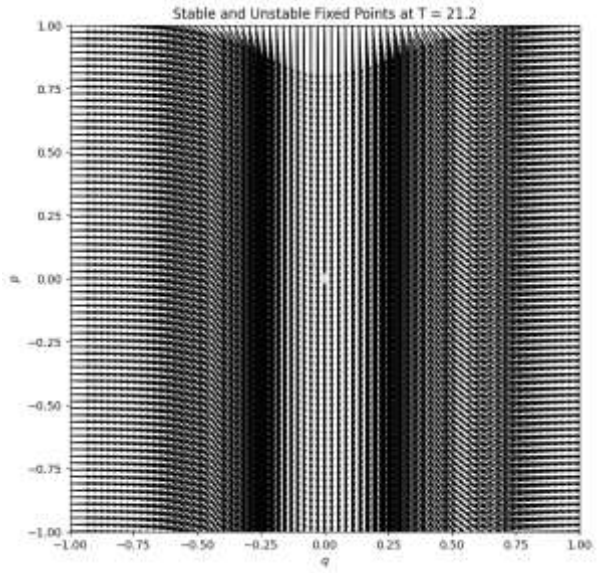


Figure 10. Project vector-field illustration at $T = 21.2$, below the reported reference critical temperature.

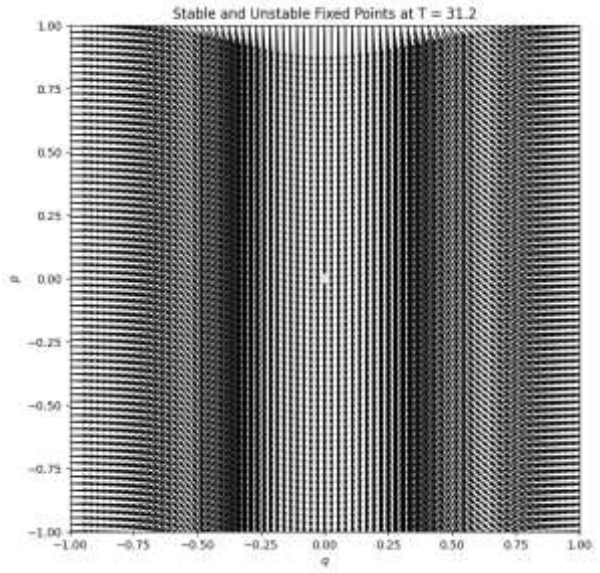


Figure 11. Project vector-field illustration at $T = 31.2$, above the reported reference critical temperature.

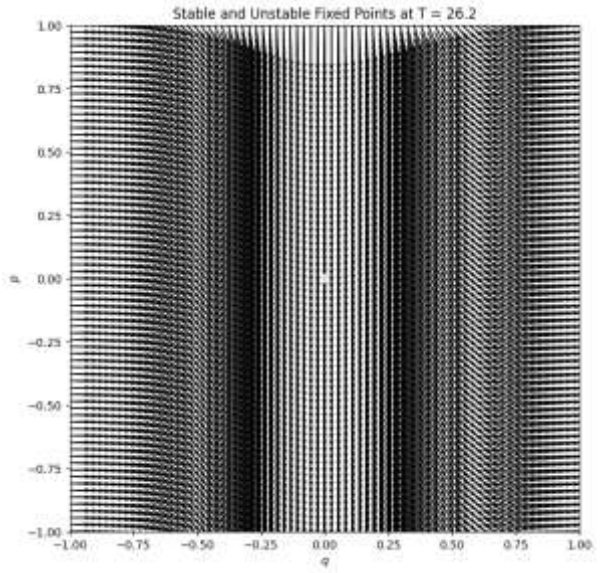


Figure 12. Project vector-field illustration at $T = 26.2$, the reported reference critical temperature.

For the scalar self-consistency equation in [1], the qualitative branch structure depends on temperature. Above the critical value, the remaining solution is the disordered state, represented by $q^* = 0$. At the threshold, an ordered and an unstable branch meet. In the coexistence regime below it, the ordered stable branch, the intermediate unstable branch, and the disordered branch are present. The unstable solution separates the attraction regions associated with the competing stable states under the chosen dynamics.

The project vector fields are illustrative. Their appearance alone does not verify local stability, and the underlying equations used to generate the arrows are not included in the supplied report. Stability should be assessed with the derivative or Jacobian of the actual iteration or dynamical system. The criterion for a discrete self-consistency iteration differs from the criterion for a continuous-time flow, so the interpretation must specify which system is being examined.

Discussion and conclusions

The finite-temperature balance model provides a statistical-mechanical description of tension in signed networks. Its mean-field formulation predicts an abrupt transition between ordered balanced configurations and a disordered branch, with competing solutions and sensitivity to initial conditions below the reported critical temperature. A balanced state may consist of two mutually antagonistic groups, so the presence of negative relationships is compatible with structural balance.

The published Monte Carlo calculations support the broad mean-field picture while also showing the importance of metastability and finite observation times [1]. The project calculations reproduce the qualitative change in the number of self-consistent solutions and the overall bifurcation structure. The energy-relaxation and temperature-scan results show differences from the published figures, particularly for random initial conditions and the apparent transition region.

These differences define the limits of the present reproduction. The available report does not establish long-time convergence or a calibrated uncertainty in the critical temperature. Matching the energy and temperature conventions, documenting the update-time normalization, extending equilibration, and comparing results across independent runs would be necessary for a quantitative assessment. The comparison with quartic balance further shows that higher-order interactions can change the number and character of stable states, and that stability must be interpreted in relation to the specific Hamiltonian and dynamical rule.

References

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2. Kargaran, A., Ebrahimi, M., Riazi, M., Hosseiny, A., and Jafari, G. R. (2020). Quartic balance theory: Global minimum with imbalanced triangles. *Physical Review E*, **102**, 012310. https://doi.org/10.1103/PhysRevE.102.012310

3. Bifurcations Part 1, Saddle-Node Bifurcation. Teaching video cited in the original report. https://www.youtube.com/watch?v=BBd68_q3Dgg

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